Table 2. **Hyperparameter details for `GPT-2 (86M)` and `SEDD (90M)` model.**

Hyperparameter	`Sibling Discovery`	`Triangle Discovery`	`Circle Construction`	`Line Construction`
Max. Learning Rate	1×10^{-4}	1×10^{-4}	1×10^{-4}	1×10^{-4}
Model Seq. Len.	128	128	128	128
Max training steps (`GPT-2 (86M)`)	$20k$	$80k$	$40k$	$40k$
Max training steps (`SEDD (90M)`)	$80k$	$80k$	$80k$	$80k$
Training size	$50k$	$15k$	$10k$	$10k$
Weight given to multi-token obj.	0.5	0.33	0.75	0.75

convergence in terms of algorithmic creativity. [1] We provide sensitive analysis of learning rate in §F. For sampling, we used the default function parameters, which turned out to involve top-K sampling with $K = 50$.

[1]We use the codebase of Lou et al. (2023) at `https://github.com/louaaron/Score-Entropy-Discrete-Diffusion`.

E. Sensitivity analyses for `Gemma v1 (2B)`

In this section, we report that our observations are robust to the choice of various hyper-parameters. First, we present a series of plots for the `Gemma v1 (2B)` model; each group of plots reports varying one hyperparameter for all the datasets. Fig 11 for train set size, Fig 12 for task complexity, Fig 13 for the weight given to the multi-token objective (and Fig 14 correspondingly for memorization), Fig 15 for learning rates, Fig 16 for number of training steps and Fig 17 for batch size. In §E.1, we report analyses for varying sampling conditions. It is worth noting that the occasional exceptions to our trends generally come from `Line Construction`, suggesting that this task is most friendly towards next-token prediction of the four we study.

Note on task-complexity. In Fig 12, we report robustness of our results to variations in the task complexity (e.g., degree, path length etc.,). Note that the variations we have explored are within reasonable factors. If we vastly increase certain factors (e.g., increase the degree of the vertices), we expect learning to become either highly trivial or non-trivial (see §C for some reasoning). Besides, as discussed in the main paper, teacherless training is a hard objective to optimize especially for smaller models; thus, we expect increasing the task complexity beyond a point to hurt the teacherless model for a fixed model size (crucially, for optimization reasons, not generalization reasons).

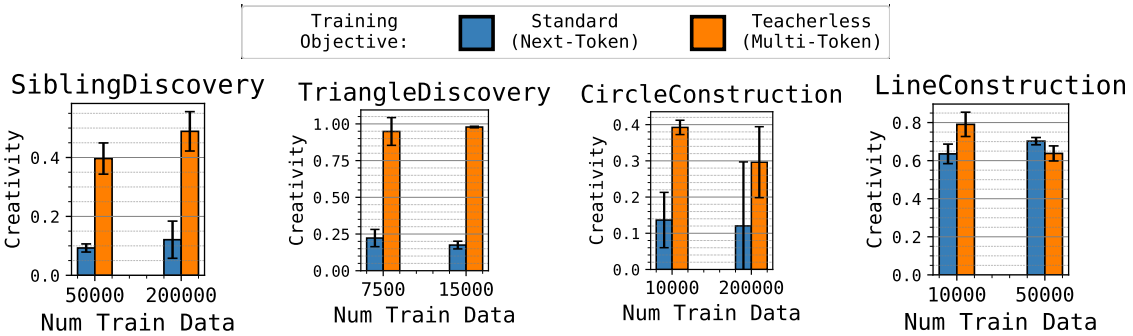

Figure 11. **Training size and algorithmic creativity for `Gemma v1 (2B)`:** Algorithmic creativity increases under multi-token prediction across various training set sizes. Note though that, in our examples, we except the gap to diminish eventually with sufficiently many training datapoints (this is unlike the failure of next-token prediction in B&N'24).

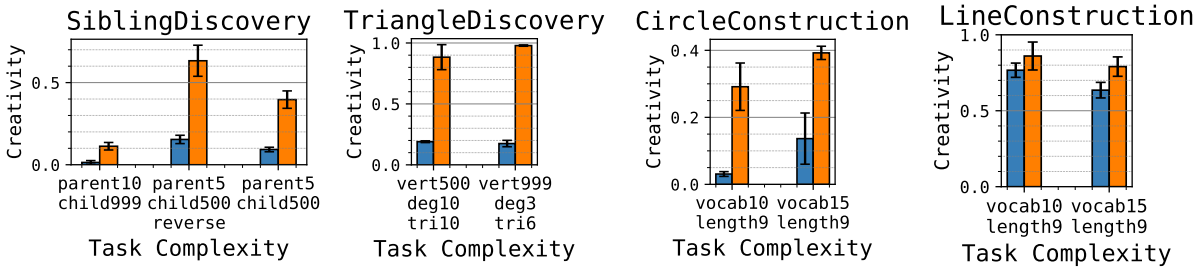

Figure 12. **Task complexity and algorithmic creativity for `Gemma v1 (2B)`:** Algorithmic creativity increases under multi-token prediction across (reasonable) variations in the dataset parameters (as described in §C).

E.1. Varying sampling methods

Fig 18, Fig 19, and Fig 20 report creativity, memorization and coherence (i.e., fraction of generated strings that are coherent) for various sampling methods (greedy decoding and nucleus sampling) with various prefix conditionings. Here we not only look at the baseline (denoted by `null`) and seed prefix (denoted by `seed`), we also look at using a prefix of pause tokens (Goyal et al., 2024) (denoted by `pause`). This can be viewed as way to provide extra computation to the model before it emits its outputs. If this had the same effect of seed-conditioning, then we would conclude that the gains of seed-conditioning come purely from prepending extra computation to the model — not from the act of injecting noise at the input. However, in our experiments, we do not find noticeable gains from pause-conditioning, establishing that noise-injection is in itself beneficial.

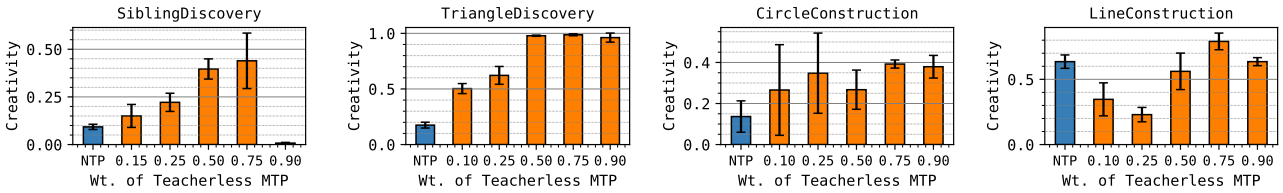

Figure 13. **Weight given to multi-token objective and algorithmic creativity for `Gemma v1 (2B)`:** Algorithmic creativity increases under multi-token prediction across various weights given to the multi-token component of the objective, barring some deviations for `Line Construction`.

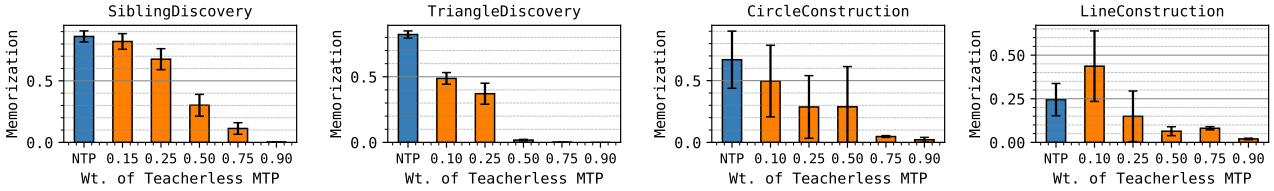

Figure 14. **Weight given to multi-token objective and memorization score for `Gemma v1 (2B)`:** Memorization reduces under multi-token prediction across various weights given to the multi-token component of the objective.

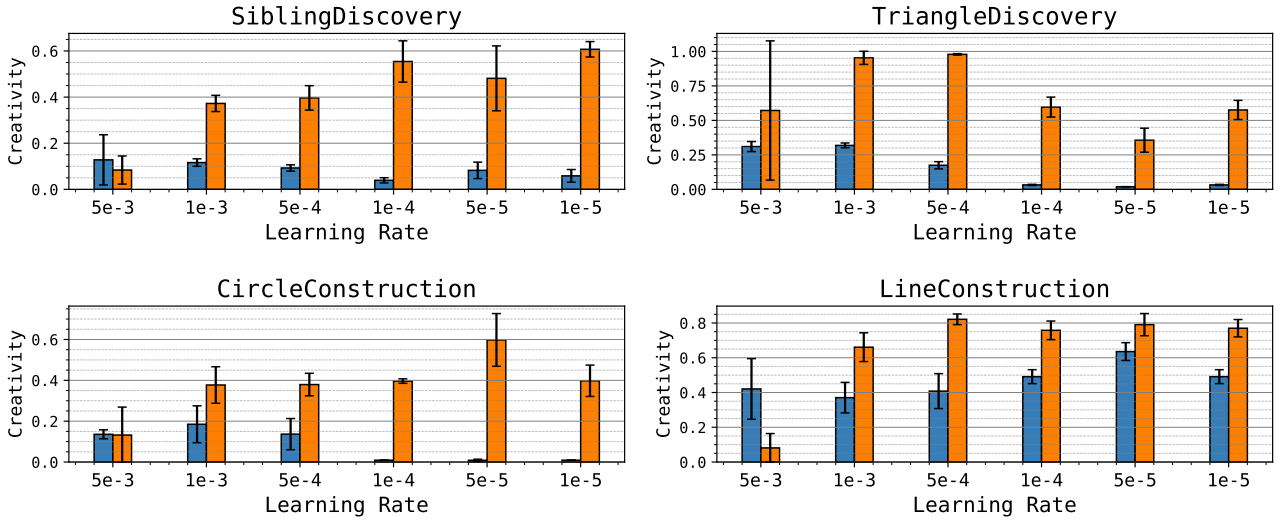

Figure 15. **Learning rate and algorithmic creativity for `Gemma v1 (2B)`:** Algorithmic creativity increases under multi-token prediction across various learning rates.

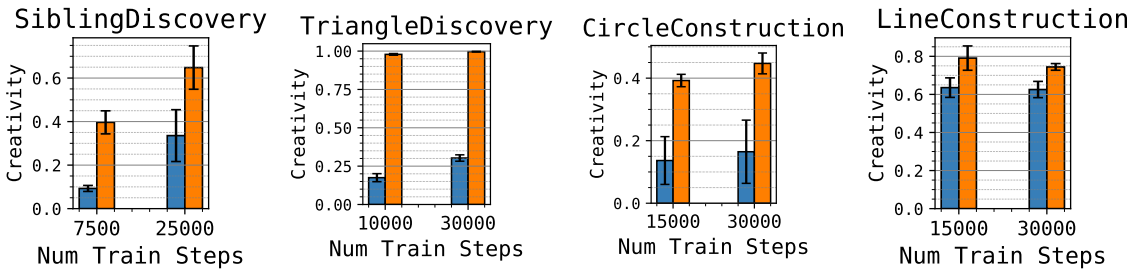

Figure 16. **Training steps and algorithmic creativity for `Gemma v1 (2B)`:** Algorithmic creativity under multi-token prediction across lengths of training.